25.01.2024

Elliptik egri chiziqlar.

AQSH 2000-y: DSA ERI → ECDSA - ERI

RF 2001-y: ГОСТР34.10-94 → ГОСТР34.10-2001.

DH protokol → ECDH.

Özbekiston.

— ÖzDSt 1092:2009 – ERI st.

- 1-alg: Parametrli algebraga asoslangan ERI (Chekli maydonda diskret logarifmlashga asoslangan).
- 2-alg: ГОСТ Р34.10-2001

— ÖzDSt 2826:2014 – ERI (EECH ga asoslangan, parametrli algebraga asoslangan).

Elliptik egri chiziqlar, uning grafiklari, tegishli nuqtalarni qoshish formulalari.

Ta'rif: Biror K-chekli maydonda olingan elliptik egri chiziq deb, Veyershtrass tenglamasi quyidagi

$$y^2+a_1xy+a_3y=x^3+a_2x^2+a_4x+a_6$$

tenglik orqali aniqlanuvchi egri chiziqqa aytiladi, bu yerda $a_1, a_2, \dots, a_6 \in$

~~Ta'rif: Ushbu E: f(x,y)~~

Bugungi kunda kriptografiyada 2 xil turdagi elliptik egri chiziqlar keng qollaniladi:

1) $y^2 = x^3+ax^2+bx+c \pmod p$
2) $y^2 = x^3+ax+b \pmod p$

29.01.2024.

EECh nuqtalarini aniqlash

EECh nuqtalarini qoshish formulalari

Berilgan EECh uchun uning nuqtalarini topishning 2 ta usuli mavjud: $(E: y^2=x^3+ax^2+bx+c$; (1) $P(x,y)$, $G(x,y) \in$

1-usul: Tanlangan $y^2=x^3+ax+b$ $(a,x,y \in R)$ tenglamaga x_i qiymatlar berib, tenglamaning ong tomoni tola kvadrat tashkil qilishi tekshiriladi. Agar qandaydir x_k qiymatda tola kvadrat tashkil qilsa, u holda o'zgaruvchi $x = x_k$, $y = \sqrt{x_k^3 + ax_k + b}$ teng bo'ladi.

Misol:

$a=2$, $b=-3$ bo'lsin, $x=2$ bo'lganda

$y = \sqrt{2^3 + 2\cdot 2 - 3} = 3$

$P(2, 3)$ ratsional nuqtaga ega.

2-usul:

Tenglamaning a-koeffitsiyenti va (x, y) nuqtani fiksirlab $(a, x, y \in R)$ $b = y^2 - x^3 - ax$ formula orqali b-koeffitsiyent hisoblab topiladi.

Misol: $a=2$, $(x, y) = (1, 2)$ bo'lsin.

U holda $b = 2^2 - 1^3 - 2\cdot 1 = 1$

Demak $(1, 2)$ nuqta ($y^2 = x^3 + ax + b$) $y^2 = x^3 + 2x + 1$ ga tegishli ratsional nuqta hisoblanadi.

Vazifa

1) $x^2 - y^2 = 1$ tenglamaning barcha ratsional yechimlarini toping.

$(x, y) = \left(\frac{a}{b}, \frac{c}{d}\right)$, $a, b, c, d \in Z$.

2) $y(6-y) = x^3 - x$

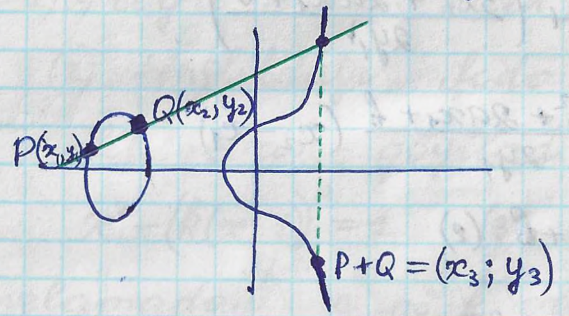
$\exists (x, y) = \left(\frac{a}{b}, \frac{c}{d}\right)$

3) $x^2 + y^2 = 3$

yoki $x^2 + 82y^2 = 3$

Faraz qilaylik $\mathcal{E}: y^2 = x^3 + ax^2 + bx + c$ (1)

$\forall P, Q \in \mathcal{E}$ nuqtalar $\mathcal{E}$ECh ga tegishli.



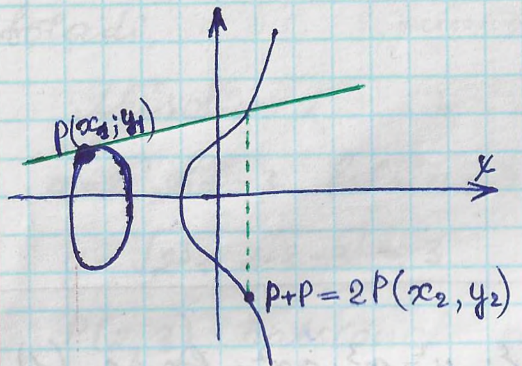
$y^2 = x^3 + ax^2 + bx + c$ (1)

$$\begin{cases} x_3 = \left(\frac{y_2 - y_1}{x_2 - x_1}\right)^2 - (a + x_1 + x_2). \\ y_3 = -y_1 + \left(\frac{y_2 - y_1}{x_2 - x_1}\right)(x_1 - x_3) \end{cases}$$

$y^2 = x^3 + ax + b$ (2)

$$\begin{cases} x_3 = \left(\frac{y_2 - y_1}{x_2 - x_1}\right)^2 - x_1 - x_2 \\ y_3 = -y_1 + \left(\frac{y_2 - y_1}{x_2 - x_1}\right)(x_1 - x_3) \end{cases}$$

ΕΕCh da P+P nuqtani hisoblash.



$$y^2 = x^3 + ax^2 + bx + c \quad (1)$$

$$\begin{cases} x_2 = -2x_1 - a + \left(\dfrac{3x_1^2 + 2ax_1 + b}{2y_1}\right)^2 \\ y_2 = -y_1 - \dfrac{3x_1^2 + 2ax_1 + b}{2y_1}(x_2 - x_1) \end{cases}$$

$$y^2 = x^3 + ax + b \quad (2)$$

$$\begin{cases} x_2 = \left(\dfrac{3x_1^2 + a}{2y_1}\right)^2 - 2x_1 \\ y_2 = -y_1 + \dfrac{3x_1^2 + a}{2y_1}(x_1 - x_2) \end{cases}$$

$$\left[\begin{array}{l} 3P = P + 2P \\ 4P = P + 3P \\ 5P = 4P + P \ \dots \end{array}\right]$$

Tasdiq: Agar $P, Q \in \mathcal{E}$ nuqtalar ratsional nuqtalar bolsa, u holda $R(x_3, y_3)$ nuqta ham ratsional nuqta boladi.

Vazifa

1) $x^2 - y^2 = 1$ (1) tenglamaning ratsional yechimlarini toping.

$(x;y) = (\frac{a}{b}; \frac{c}{d})$, $\quad a, b, c, d \in Z$

Yechish: $(1;0)$ yechim ushbu tenglamaning yechimi ekanligi yaqqol kорinib turibdi. Shuning uchun (1) tenglama $y = k(x-1)$ (2) togri chiziq bilan kesishishi aniq. (2) tenglikdagi ifodani (1) tenglamaga olib borib qoyiladi.

$$x^2 - (k(x-1))^2 = 1 \qquad (3)$$

(3) tenglamadan x ni k orqali ifodalaymiz (tenglamani yechamiz).

$$x^2 - k^2(x^2 - 2x + 1) = 1$$
$$x^2 - k^2x^2 + 2k^2x - k^2 = 1$$
$$(1-k^2)x^2 + 2k^2x - (k^2+1) = 0$$
$$D = 4k^4 + 4(k^2+1)(1-k^2) = 4k^4 + 4 - 4k^4 = 4.$$
$$x_1 = \frac{-b - \sqrt{D}}{2a} = \frac{-2k^2 - 2}{2(1-k^2)} = \frac{k^2+1}{k^2-1}$$
$$x_2 = \frac{-b + \sqrt{D}}{2a} = \frac{-2k^2 + 2}{2(1-k^2)} = 1.$$

$x=\frac{k^2+1}{k^2-1}$ ni (1) yoki (2) tenglamaga qoʻyib y ni topamiz.

$$y=k(x-1)=k\cdot\left(\frac{k^2+1}{k^2-1}-1\right)=k\left(\frac{k^2+1-k^2+1}{k^2-1}\right)=$$
$$=\frac{2k}{k^2-1}.$$

Javob: $(x;y)=(1;0)$ va $\left(\frac{k^2+1}{k^2-1};\frac{2k}{k^2-1}\right)$
$k\neq\pm1$

2) $y(6-y)=x^3-x$ (1)

$x=3y-1$ ni tanlab (1) tenglamadagi x ni oʻrniga qoʻyamiz: (maqsad: y^1 ni tenglamadan yoʻqotish).

$$y(6-y)=(3y-1)\left((3y-1)^2-1\right)$$
$$6y-y^2=(3y-1)(9y^2-6y)$$
$$\cancel{6y}-y^2=27y^3-9y^2-18y^2+\cancel{6y}$$
$$27y^3-26y^2=0$$
$$y^2(27y-26)=0$$

$y=0 \to x=-1;0;1$

$y=\frac{26}{27} \to x=\frac{17}{9}$

Javob: $(-1;0),(0;0),(1,0)$ $\left(\frac{17}{9};\frac{26}{27}\right)$

31.01.2024.

Elliptik egri chiziq nuqtalari tartibi

$$[n]P=P+P+\dots+P.$$

Mazkur qoshish jarayonida quyidagi 2 ta holat roʻy berishi mumkin:

1) Biror n-qadamda $[n]P=0$ tenglik bajarilishi mumkin.

2) $2P$, $3P$, $4P$ va hokazo $[n]P$-nuqtalar har xil qiymatga ega boʻlishi mumkin.

Ta'rif: Agar $mP\neq0$, barcha $m<n$, bajarilib, $[n]P=0$ boʻlsa, u holda P-nuqta n-chekli tartibga ega deyiladi.

Misol.

$y^2=x^3+4$, $P(0,2)$ nuqta $n=3$ tartibli ekanligi koʻrsatilsin.

Yechish: $a=b=0$

$x_2=0$, $y_2=-2$.

$x_1=x_2 \to$ demak $n=3$

Bevosita tekshirib ko'rish mumkinki

1) $y^2=x^3+1$, $P(2,3)$, $n=6$

2) $y^2=x^3-43x+166$, $P(3,8)$, $n=7$.

Vazifa

2) $y^2=x^3-43x+166$ $\quad a=-43 \quad b=166 \quad P(3,8)$

2P) $x_2=\left(\frac{3x_1^2+a}{2y_1}\right)^2-2x_1=\left(\frac{3\cdot 3^2+(-43)}{2\cdot 8}\right)^2-2\cdot 3=$

$=\left(\frac{27-43}{16}\right)^2-6=-5$

$y_2=-y_1+\frac{3x_1^2+a}{2y_1}(x_1-x_2)=-8+\frac{3\cdot 3^2-43}{2\cdot 8}\cdot(3-(-5))=$

$=-8+(-1)\cdot(8)=-16$

~~$2P \Rightarrow (-7;-18)$~~ $\quad 2P \Rightarrow (-5;-16)$

3P) $x_3=\left(\frac{y_2-y_1}{x_2-x_1}\right)^2-x_1-x_2=\left(\frac{-16-8}{-5-3}\right)^2-(3+(-5))=$

$\cancel{=\frac{169}{25}+4=}$ $=9+2=11$

~~$3P \Rightarrow$~~

$y_3=-y_1+\left(\frac{y_2-y_1}{x_2-x_1}\right)(x_1-x_3)=-8+\left(\frac{-16-8}{-5-3}\right)(3-11)=$

$=-8+(-24)=-32$

$3P \Rightarrow (11;-32)$

4P) $x_4=\left(\frac{y_3-y_1}{x_3-x_1}\right)^2-x_1-x_3=\left(\frac{-32-8}{11-3}\right)^2-3-11$

$=25-14=11.$

$y_4=-y_1+\left(\frac{y_3-y_1}{x_3-x_1}\right)(x_1-x_4)=-8+\left(\frac{-32-8}{11-3}\right)\cdot(3$

$=-8+40=32$

$4P \Rightarrow (11;32)$

5P) $x_5=\left(\frac{y_4-y_1}{x_4-x_1}\right)^2-x_1-x_4=\left(\frac{32-8}{11-3}\right)^2-3-11=$

$=9-14=-5$

$y_5=-y_1+\left(\frac{y_4-y_1}{x_4-x_1}\right)(x_1-x_5)=-8+\left(\frac{32-8}{11-3}\right)(3-(-5))$

$=-8+24=16$

$5P \Rightarrow (-5;16)$

6P) $x_6=\left(\frac{y_5-y_1}{x_5-x_1}\right)^2-x_1-x_5=\left(\frac{16-8}{-5-3}\right)^2-3-(-5)=$

$=1-3+5=3$

$y_6=-y_1+\left(\frac{y_5-y_1}{x_5-x_1}\right)(x_1-x_6)=-8+\left(\frac{16-8}{-5-3}\right)(3-3)=$

$6P \Rightarrow (3;-8)$ $\quad x_6=x_1$

Javob: $n=7$.